LESSON 8.9
NETS OF RECTANGULAR PRISMS AND PYRAMIDS

LESSON INTRODUCTION

MA.6.GR.2.4 Given a mathematical or real-world context, find the surface area of right rectangular prisms and right rectangular pyramids using the figure's nets.

LEARNING TARGETS

- I can draw the net of a rectangular prism or rectangular pyramid.
- I can relate a net to its three-dimensional shape.
- I can relate a rectangular prism or rectangular pyramid to its net.

BENCHMARK COHERENCE

BUILDING ON

N/A

WORKING TOWARD

MA.912.GR.4.6 Solve mathematical and real-world problems involving the surface area of three-dimensional figures limited to cylinders, pyramids, prisms, cones and spheres.

ESSENTIAL QUESTIONS

- What is a net?
- What are the similarities or differences between a net of a three-dimensional figure and a composite figure?
- What are some differences between the nets of rectangular prisms and the nets of rectangular pyramids?

LESSON NARRATIVE

This lesson begins with a review of the previous lesson finding a missing dimension when given volume. The lesson begins with a review on composite figures and relates the composite figures to the net of a three-dimensional figure. Students explore how nets represent the two-dimensional faces of the three-dimensional figures and collaboratively cut out, fold, and create prisms to match three-dimensional solids to the nets that they form when folded. The lesson wraps up with students' self-reflections on their learning about nets.

COMPONENT SUMMARY

8.9.1 WARM-UP *(~5 MIN.)*

Review finding missing dimension of prism when given volume

8.9.2 REFRESHER *(10 MIN.)*

Review how to find the total area of composite figures

8.9.3 EXPLORATION *(5-10 MIN.)*

How three-dimensional figures relate to given nets

8.9.4 COLLABORATION *(10-15 MIN.)*

Draw and label nets for the given three-dimensional figures

8.9.5 WRAP-UP *(~5 MIN.)*

Reflect on the most difficult part of the lesson

RESOURCES

GLOSSARY

Key Terms:

- Net

Additional Terms:

- Face
- Cube

MATERIALS

- Nets for 8.9.3 Exploration
- Scissors
- Four-function calculator
- (Optional) Tape for constructing nets

ADDITIONAL PREPARATION

- Print and cut out nets for 8.9.3 Exploration
- (Optional) Gather real-world examples of nets and models of three-dimensional solids.

TEACHER TIPS

COMMON MISCONCEPTIONS

- Students may struggle with labeling all the sides or faces in a net.
- Students may struggle recognizing or finding missing side lengths.
- Students may struggle with visualizing the three-dimensional solid that the nets create.
- Students may struggle with connecting different nets to the same three-dimensional figure.

THINGS TO CONSIDER

- Students should be monitored closely during the 8.9.3 Exploration to be sure that they cut and fold the nets to correctly form the intended solid. Having examples for students to refer to may be helpful.
- Teacher scaffolding through the 8.9.2 Refresher may be beneficial to help students build connections between two-dimensional composite figures and the prisms they represent.
- Consider gathering or having students recognize real-world examples of nets (e.g., wrapping a box in gift paper or unfolding a shoe box).

LITERACY AND LANGUAGE ROUTINES

Compare and Connect

Like Lesson 8, this lesson provides important opportunities for students to compare and connect concepts related to representing three-dimensional figures and their characteristics. Consider fostering discussion or using a Venn diagram to organize and record student thinking during 8.9.3 Exploration as students compare the idea of a composite figure and a net of a three-dimensional figure.

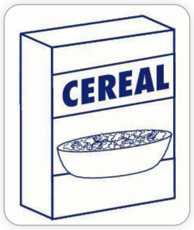
MATHEMATICAL THINKING AND REASONING (MTR) STANDARD

MA.K12.MTR.2.1 Multiple Representations

As students connect two-dimensional nets to three-dimensional prisms or pyramids, they are deepening their understanding of how to represent shapes in multiple ways. Clarifications within this standard emphasize connections between concept and representation. Utilize the progression of the lesson to highlight how nets create three-dimensional solids, including by physically cutting out and building prisms from printed nets.

DIFFERENTIATE INSTRUCTION

Consider extending the lesson by having students unfold and refold real-life examples of three-dimensional figures, such as shoe boxes, after completing 8.9.3 Exploration. Allowing students physical models of the shapes may strengthen the connection between the nets and the solids for learners.



8.9.1 WARM-UP: BELL WORK

Component Overview: *Students are given the volume, length, and depth of a real-world three-dimensional shape and find the missing width. Have students work individually and consider using responses as an additional data point or formative assessment of learning targets from Lesson 8.*

DIFFERENTIATE INSTRUCTION

Pictures or drawn models of the scenario can provide clarity and strengthen understanding. Encourage the students to draw a labeled picture of the figure described in question 1 prior to solving the problem and to label the length, depth, volume, and eventually the width as they work.

8.9.1 Warm-Up: Bell Work

1. A rectangular ball pit has a volume of 28 cubic meters (m^3). Find the width of the ball pit if the length is $5\frac{5}{6}$ m and the depth is $\frac{3}{4}$ m.

$$28\ m^3 = \frac{35}{6}\text{m} \times w \times \frac{3}{4}\text{m}$$

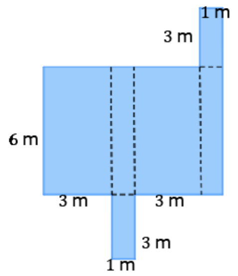
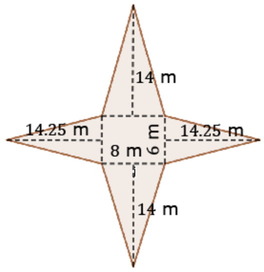
$$28\ m^3 = \frac{105}{24}m^2 \times w$$

$$w = \frac{28\ m^3}{\frac{105}{24}m^2} = \frac{32}{5}\text{m}$$

$$w = 6\frac{2}{5}\text{m}$$

8.9.2 Refresher: Area of Composite Figures

1. Work with your partner to find the area of each composite figure. All measurements are in meters (m).

<table>
<tr><th>Figure</th><th>Area</th></tr>
<tr><td>All angles are right angles.

</td><td>$54\ m^2$</td></tr>
<tr><td>All dashed lines intersect at right angles.

</td><td>$245.5\ m^2$</td></tr>
</table>

8.9.2 REFRESHER: AREA OF COMPOSITE FIGURES

Component Overview: *This section reviews how to find the total area of composite figures. Students find missing dimensions and determine the combined area of various rectangles and triangles, and though it is not yet connected, students are finding surface area of a rectangular prism and pyramid. As students work in partners, encourage students to label all missing dimensions prior to finding the area and have students use formulas to find the area. Consider working with a small group of students to scaffold review of area for any students who need additional support.*

ENRICHMENT

To support connecting these composite figures to the nets of rectangular and triangular prisms, pose these questions:

- What shapes are included when finding the total area of the first figure?
- What shapes are included when finding the total area of the second figure?

TEACHER MOVES

Compare and Connect

Compare and contrast the similarities and differences in finding the area of each of the composite figures. Students may point out that the first figure is made up of six shapes and the second figure is made up of fives shapes. Students may also respond that the second shape appears to have a shape in the center that is a different shape than the shapes surrounding it.

8.9.3 EXPLORATION: CREATING THREE-DIMENSIONAL FIGURES

Component Overview: *Students discover which nets represent which three-dimensional figure: a rectangular prism and a pyramid. Students also make connections to composite figures and nets in questions 3 and 4. Consider explaining question 1 together as a class, then prompting students to work in partners or small groups as they complete the table in question 2. Be sure to provide enough copies of nets and scissors to each group and encourage students to fold and connect the edges of the net to see the prism or pyramid come to life through folding.*

ENRICHMENT

Allow students to answer these questions as they explore to strengthen the connection from composite figures to the nets:

- What is similar about a net of a three-dimensional figure and a composite figure?
- What are some real-world examples of nets?

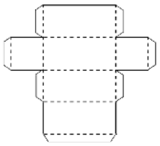
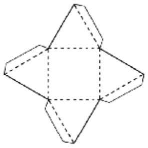
8.9.3 Exploration: Creating Three-Dimensional Figures

Both composite figures in the previous component are nets. A net is formed when the faces of a three-dimensional figure are positioned by arranging the faces to create a two-dimensional composite figure.

A ***net*** is a two-dimensional diagram that can be folded or made into a three-dimensional figure.

1. **Your teacher will provide you with copies of some nets. Cut the nets along the solid line. Fold the nets along the dotted line and use the tabs to glue the figure together.**

2. **Complete the table to describe the three-dimensional figure formed.**

Net	Three-Dimensional Figure Formed
	*Rectangular Prism*
	*Pyramid*

3. **Look back at the two composite figures from the refresher. Predict what three-dimensional figure is made from cutting out and folding up the sides of those figures.**
   Rectangular Prism, Pyramid

4. **With your partner, discuss how the area of a two-dimensional composite figure is related to the area of the faces of the three-dimensional figure. Summarize your discussion.**
   The area of a 2-dimensional composite shape will help you find the area of the outside of a 3-dimensional shape.

8.9.3 EXPLORATION: CREATING THREE-DIMENSIONAL FIGURES (CONTINUED)

TEACHER MOVES

Compare and Connect

At the end of the Exploration, regroup students as a class to discuss responses to questions 3 and 4. Position students to make connections between composite figures as the faces of three-dimensional figures. Use real-world examples, like wrapping a box in paper, to solidify that the area of the two-dimensional shape is the same as the area of the outside of the three-dimensional shape. Ask:

- What shapes are the faces of a pyramid?
- What shapes are the faces of a prism?

8.9.4 COLLABORATION: DETERMINING NETS

Component Overview: *Students collaborate by sketching and labeling nets for two three-dimensional figures: a cube and a rectangular pyramid. Students build on previous knowledge by identifying dimensions of the figures. Before students begin the collaboration, discuss ideas for the individual two-dimensional shapes that make up the faces of each three-dimensional figure. Encourage students to be sure that the individual shapes, or faces, are all connected so that none of the shapes are drawn by themselves.*

DIFFERENTIATE INSTRUCTION

Prior to starting the 8.9.4 Collaboration, have students match the nets made from 8.9.3 Exploration with the solids in the table. Though the dimensions are different, students may benefit from visually representing the same shape as a two-dimensional collection of shapes as they draw the nets in question 1. Also, consider providing additional examples of drawings of the shapes or physical models and providing different nets that match to the same three-dimensional figure.

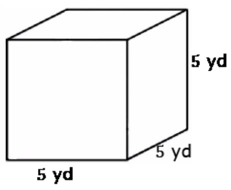
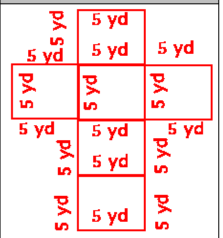
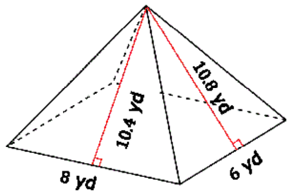
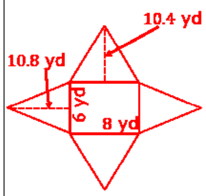
ENRICHMENT

Reinforce matching dimensions on different kinds of rectangular prisms by asking the students the following questions:

- How were you able to calculate all the missing sides for the prism and pyramid in the table?
- How can you tell the figure in the table is a cube? What shape are the faces of a cube?
- How are cubes and rectangular prisms the same?

8.9.4 Collaboration: Determining Nets

1. **With your partner, sketch the net or sketch the individual faces of the three-dimensional figure. Label each side of the net with the appropriate measurement in yards (yd).**

Three-Dimensional Figure	Net
	
	

8.9.5 Wrap-Up: Reflection

1. What was the ***muddiest*** (most difficult or confusing) point in today's lesson on nets?

Answers will vary. The muddiest thing about today's lesson for me was trying to visualize what the sides of the 3-dimensional shape looked like. The muddiest thing about today's lesson for me was trying to visualize what a shape on paper looks like folded together.

8.9.5 WRAP-UP: REFLECTION

Component Overview: *Students reflect on the lesson to determine the most difficult or confusing point on the lesson about nets. Consider collecting these responses to address misconceptions or reinforce challenging concepts at the close of the lesson.*

DIFFERENTIATE INSTRUCTION

Consider asking students to complete a table where they will identify the clearest point of the lesson and one unclear point of the lesson as well as the muddiest point of the lesson. By encouraging students to think about a range of understanding, students may avoid responding that the entire lesson was "muddy."

TEACHER MOVES

Consider prompting students to provide evidence for their responses. Providing sentence stems for them to complete may assist with communicating their thoughts clearly, such as:

- "The muddiest point was_____ because_________."
- "I found the point _______ confusing because I did not understand _________________________."